Priority Scheduling

Q1.Write a C program to simulate the Priority Scheduling (Non preemptive).

OS LAB > C P3.c > main()

```c
#include<stdio.h>
int main()
{
    int n,i,time=0,done=0,highest;
    int at[10],bt[10],pr[10],ct[10],tat[10],wt[10],completed[10]={0};
    printf("Enter number of processes: ");
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("P%d Arrival Time: ",i+1);
        scanf("%d",&at[i]);
        printf("P%d Burst Time: ",i+1);
        scanf("%d",&bt[i]);
        printf("P%d Priority: ",i+1);
        scanf("%d",&pr[i]);
    }
    while(done<n)
    {
        highest=-1;
        for(i=0;i<n;i++)
        {
            if(at[i]<=time&&completed[i]==0)
            {
                if(highest==-1||pr[i]<pr[highest])
                    highest=i;
            }
        }
        if(highest==-1)
        {
            time++;
            continue;
        }
        time+=bt[highest];
        ct[highest]=time;
        completed[highest]=1;
        done++;
    }
    for(i=0;i<n;i++)
```


```c
        }
        if(highest==-1)
        {
            time++;
            continue;
        }
        time+=bt[highest];
        ct[highest]=time;
        completed[highest]=1;
        done++;
    }
    for(i=0;i<n;i++)
```


```c
    {
        tat[i]=ct[i]-at[i];
        wt[i]=tat[i]-bt[i];
    }
    printf("\nProcess\tAT\tBT\tPR\tCT\tTAT\tWT\n");
    for(i=0;i<n;i++)
        printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",i+1,at[i],bt[i],pr[i],ct[i],tat[i],wt[i]);
}
```

```
cd "/Users/sharan/Documents/C/OS LAB/" && gcc P3.c -o P3 && "/Users/sharan/Documents/C/OS LAB/"P3
→  C cd "/Users/sharan/Documents/C/OS LAB/" && gcc P3.c -o P3 && "/Users/sharan/Documents/C/OS LAB/"P3
Enter number of processes: 5
P1 Arrival Time: 0
P1 Burst Time: 8
P1 Priority: 1
P2 Arrival Time: 1
P2 Burst Time: 1
P2 Priority: 2
P3 Arrival Time: 2
P3 Burst Time: 3
P3 Priority: 3
P4 Arrival Time: 3
P4 Burst Time: 2
P4 Priority: 4
P5 Arrival Time: 4
P5 Burst Time: 5
P5 Priority: 5

Process AT      BT      PR      CT      TAT     WT
P1      0       8       1       8       8       0
P2      1       1       2       9       8       7
P3      2       3       3       12      10      7
P4      3       2       4       14      11      9
P5      4       5       5       19      15      10
→  OS LAB
```

Priority Scheduling

Q2.Write a C program to simulate the Priority Scheduling (Non preemptive).

OS LAB > C P4.c > main()

```c
#include<stdio.h>
int main()
{
    int n,i,time=0,remain,highest;
    int at[10],bt[10],rt[10],pr[10],ct[10],tat[10],wt[10];
    printf("Enter number of processes: ");
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        printf("P%d Arrival Time: ",i+1);
        scanf("%d",&at[i]);
        printf("P%d Burst Time: ",i+1);
        scanf("%d",&bt[i]);
        printf("P%d Priority: ",i+1);
        scanf("%d",&pr[i]);
        rt[i]=bt[i];
    }
    remain=n;
    while(remain!=0)
    {
        highest=-1;
        for(i=0;i<n;i++)
        {
            if(at[i]<=time&&rt[i]>0)
            {
                if(highest==-1||pr[i]<pr[highest])
                    highest=i;
            }
        }
        if(highest==-1)
        {
            time++;
            continue;
        }
        rt[highest]--;
        time++;
        if(rt[highest]==0)
        {
            ct[highest]=time;
            remain--;
        }
    }
    for(i=0;i<n;i++)
    {
        tat[i]=ct[i]-at[i];
        wt[i]=tat[i]-bt[i];
    }
    printf("\nProcess\tAT\tBT\tPR\tCT\tTAT\tWT\n");
    for(i=0;i<n;i++)
        printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",i+1,at[i],bt[i],pr[i],ct[i],tat[i],wt[i]);
}
```

OS LAB > C P4.c > main()

```c
if(highest==-1)
{
    time++;
    continue;
}
rt[highest]--;
time++;
if(rt[highest]==0)
{
    ct[highest]=time;
    remain--;
}
}
for(i=0;i<n;i++)
{
    tat[i]=ct[i]-at[i];
    wt[i]=tat[i]-bt[i];
}
printf("\nProcess\tAT\tBT\tPR\tCT\tTAT\tWT\n");
for(i=0;i<n;i++)
    printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",i+1,at[i],bt[i],pr[i],ct[i],tat[i],wt[i]);
}
```


```
cd "/Users/sharan/Documents/C/OS LAB/" && gcc P4.c -o P4 && "/Users/sharan/Documents/C/OS LAB/"P4
→  C cd "/Users/sharan/Documents/C/OS LAB/" && gcc P4.c -o P4 && "/Users/sharan/Documents/C/OS LAB/"P4
Enter number of processes: 5
P1 Arrival Time: 0
P1 Burst Time: 8
P1 Priority: 1
P2 Arrival Time: 1
P2 Burst Time: 1
P2 Priority: 2
P3 Arrival Time: 2
P3 Burst Time: 3
P3 Priority: 3
P4 Arrival Time: 3
P4 Burst Time: 2
P4 Priority: 4
P5 Arrival Time: 4
P5 Burst Time: 5
P5 Priority: 5

Process AT      BT      PR      CT      TAT     WT
P1      0       8       1       8       8       0
P2      1       1       2       9       8       7
P3      2       3       3       12      10      7
P4      3       2       4       14      11      9
P5      4       5       5       19      15      10
→  OS LAB
```